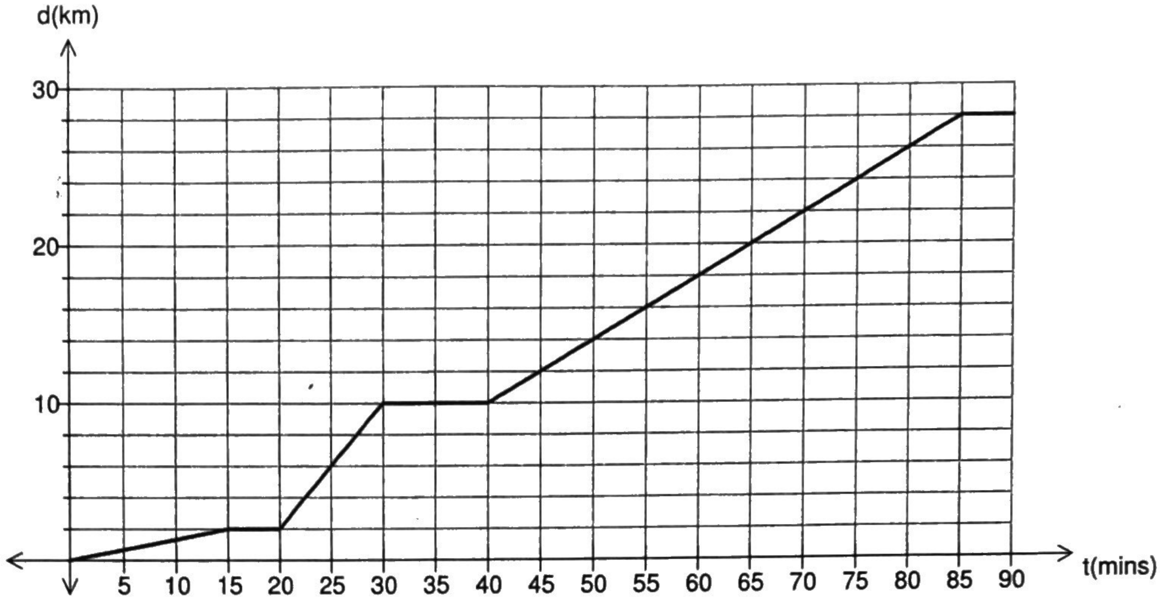
5. Ben lives in Perth and travels to Rottnest Island where he works in the hospitality industry. In order to get from his home to his work place Ben walks to the railway station, catches a train to Fremantle and then boards the 07:00 ferry bound for Rottnest Island.
The graph below represents his journey on a particular day.



(a) How long did Ben have to wait for the train to arrive on that particular day?

(b) Determine the average speed of the train in metres per second on that particular day.

(c) Determine the average speed of the ferry in kilometres per hour on that particular day.

(d) Determine the average speed for the journey from home to Rottnest Island in km/h.

(e) Ben arrived at Rottnest on this particular day. What time did Ben leave home?

In order to save the train fare Ben sometimes cycles to Fremantle.
(f) If Ben leaves home at the same time determine Ben's bicycle riding speed in metres per second if he must board the 07:00 ferry bound for Rottnest Island.

One morning Ben wakes up late and his father agrees to drive him to Fremantle to catch the 07:00 ferry. From previous experience, Ben knows that the car journey to Fremantle can be completed safely at a constant speed of 50 km/h.
(g) In order to catch the 07:00 ferry, what is the latest time that they can leave the house?

7. Sarah leaves her place of work at 5:00 pm and drives 10 km to a fresh food market to pick up fresh blueberries and other fresh produce for the weekend. Her distance from home is given by the following piece-wise equation:

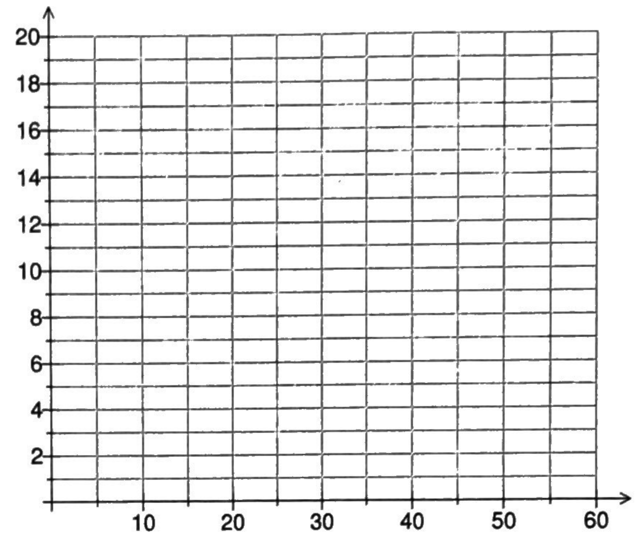
$$d = \begin{cases} t+5, & 0 \le t \le 10 \\ 15, & 10 < t \le 25 \\ 30-0.6x, & 25 < t \le 50 \end{cases}$$ where d (km) is the distance from home and t the time taken in minutes.

(a) On the axes below draw the piece-wise linear graph to represent Sarah's journey from her place of work to the markets and back to her home.

(b) How far are the markets from Sarah's home?

(c) How far is her place of work from home?

(d) Find the value of d when t = 0. What does this tell you in the context of the situation?



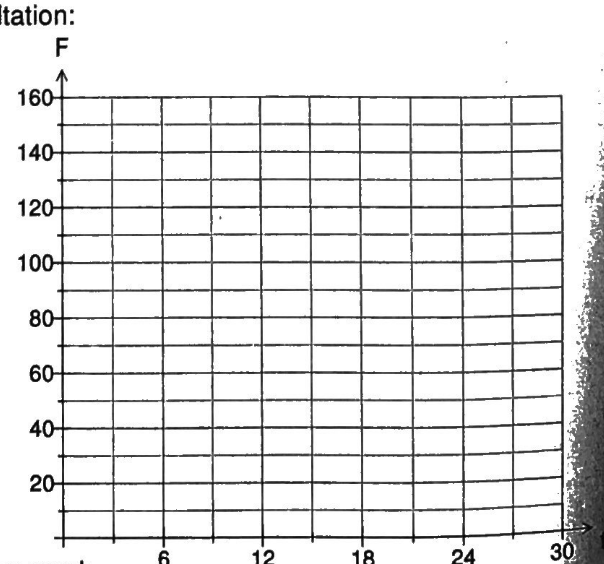
(e) What distance did Sarah travel on her way home from her workplace?

(f) What was her average speed between the markets and home?

8. A doctor's fee is based on the length of the consultation:

- up to 6 minutes costs $50
- over 6 minutes to 15 minutes costs $80
- over 15 minutes costs $80 plus $5 for every minute over 15 minutes

(a) Using the given information write a piece-wise equation for calculating the fee charged by the doctor, where F is the fee charged and t the length of the consultation.



(b) On the given axes, graph your piece-wise linear graph.

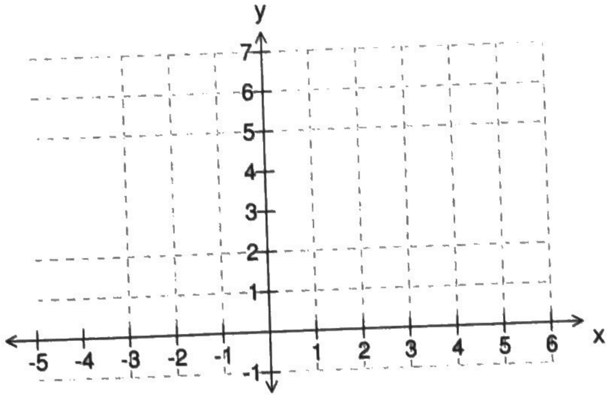
(c) How much will a 10 minute consultation cost?

(d) If the fee paid by a patient was $110, how long was the consultation?

2. (a) On the given axes sketch the graph of

$$y = \begin{cases} 1, & -4 < x \le -2 \\ 2, & -2 < x \le 0 \\ 3, & 0 < x \le 2 \\ 5, & 2 < x \le 5 \end{cases}$$

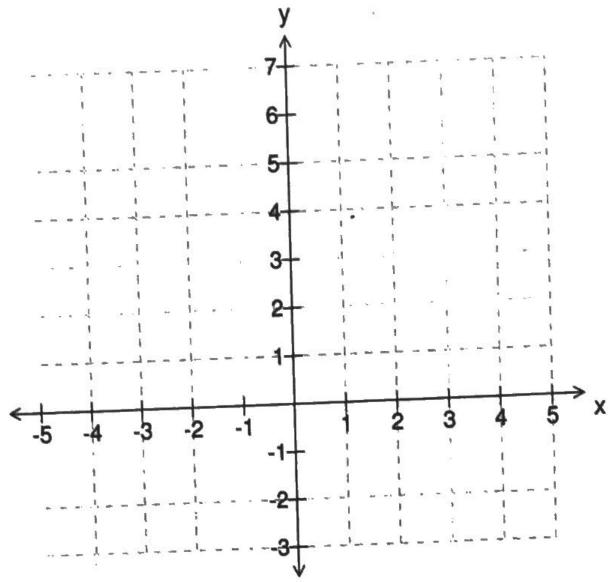
(b) Comment on the graph you have drawn.



3. (a) On the given axes sketch the graph of

$$\text{of } y = \begin{cases} 6, & -5 \le x < -3 \\ 4, & -3 \le x < -1 \\ 2, & -1 \le x < 1 \\ 0, & 1 \le x < 3 \\ -2, & 3 \le x < 5 \end{cases}$$

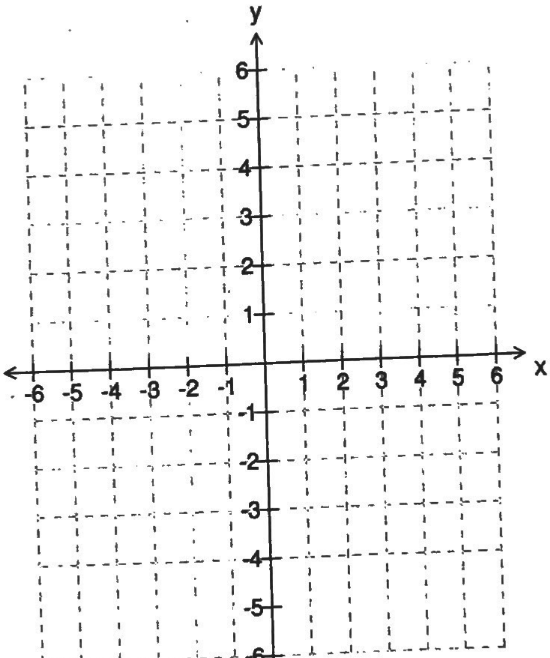
(b) Comment on the graph you have drawn.

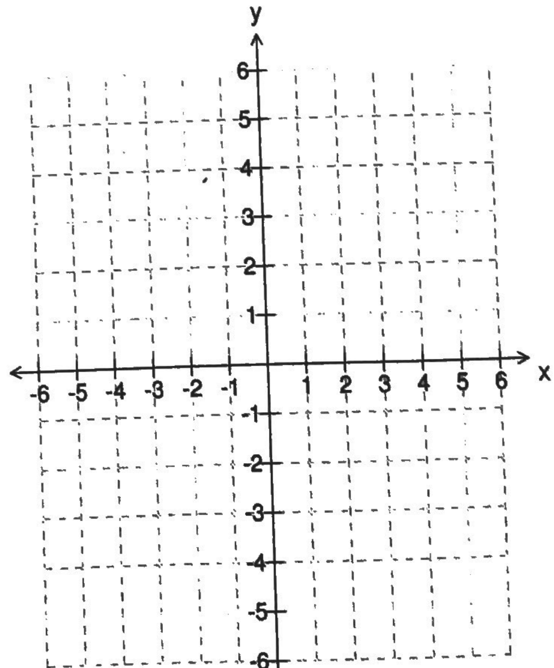


4. On the given axes draw the following piece-wise linear graphs.

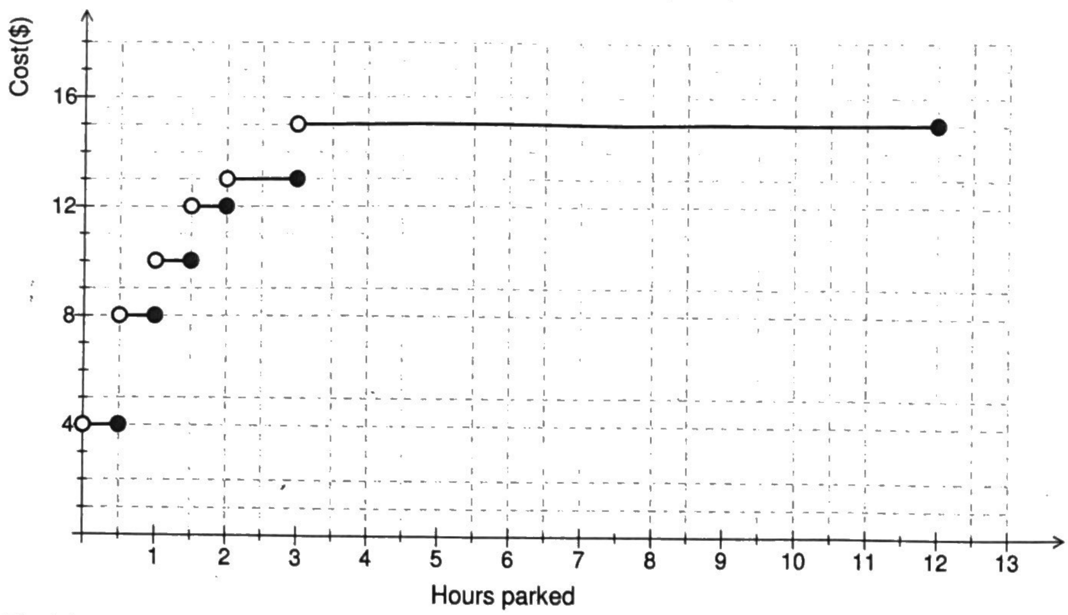
(a) $$y = \begin{cases} -4, & -5 < x \le -2 \\ -2, & -2 < x \le 1 \\ 2, & 1 < x \le 4 \\ 5, & x > 4 \end{cases}$$

(b) $$y = \begin{cases} 5, & -5 \le x < -3 \\ 2, & -3 \le x < -1 \\ -5, & -1 \le x < 1 \\ -3, & 1 \le x < 3 \\ 4, & 3 \le x < 5 \end{cases}$$





5. The graph shows the fees payable for parking at an inner city car park.



(a) Find the cost of parking for:
(i) 1.5 hours (ii) 45 minutes (iii) 5 hours (iv) 2 hours 10 minutes

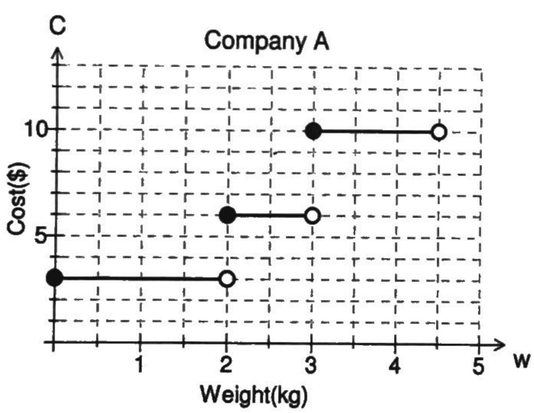
(b) What is the range of times a car can be parked for $12.

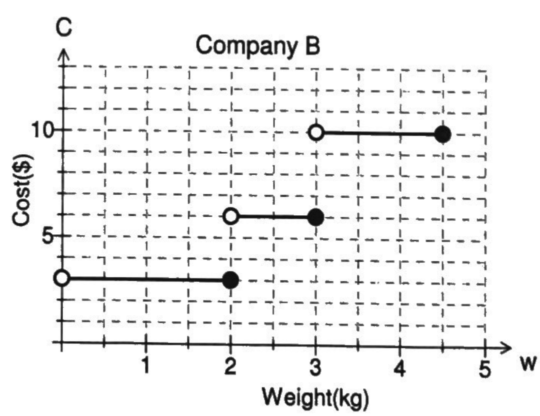
(c) How long can I park my car in this car park if I am prepared to pay no more than $10?

(d) Mel parked her car at 1:25 pm, met up with friends for coffee and left the car park at 3:08 pm. How much did she pay for parking her car?

(e) What is the maximum time that you can park in this car park and what is the charge?

6. The step graphs show the cost of sending parcels of different weights for two courier companies.





(a) List any similarities.

(b) List any differences.

(c) Which company would you use? Why? Illustrate with an example.